16

Technical data

Contents of this chapter

This chapter contains the technical specifications of the drive, for example, the ratings, sizes and technical requirements, provisions for fulfilling the requirements for CE and other markings.

Electrical ratings

This section gives ratings for the standard drive. For ratings of marine-type approved drives (option +C132), refer to ACS880-01…, ACS880-04…, ACS880-11…, ACS880-31…, ACS880-14… and ACS880-34… +C132 marine type-approved drives supplement (3AXD50000010521 [English]).

The nominal rating for the drive modules with 50 Hz and 60 Hz supply are given below.

IEC RATINGS										
ACS880-34-…	Frame size	Input current[1)]	Output ratings							
			Nominal use				Light-duty use		Heavy-duty use	
		I_1	I_{max}	I_2	P_n	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	A	A	kW	kVA	A	kW	A	kW
U_n = 400 V										
246A-3	R11	212	350	246	132	170	234	132	206	110
293A-3	R11	257	418	293	160	203	278	160	246	132
363A-3	R11	321	498	363	200	251	345	200	293	160
442A-3	R11	401	621	442	250	306	420	250	363	200
505A-3	R11	401	631	505	250	350	480	250	363	200
585A-3	R11	505	751	585	315	405	556	315	442	250
650A-3	R11	569	859	650	355	450	618	355	505	250

IEC RATINGS										
ACS880-34-...	**Frame size**	**Input current[1)]**	**Output ratings**							
			Nominal use				**Light-duty use**		**Heavy-duty use**	
		I_1	I_{max}	I_2	P_n	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	**A**	**A**	**kW**	**kVA**	**A**	**kW**	**A**	**kW**
U_n = 500 V										
240A-5	R11	169	350	240	132	208	228	132	180	110
260A-5	R11	205	418	260	160	225	247	160	240	132
361A-5	R11	257	542	361	200	313	343	200	260	160
414A-5	R11	321	614	414	250	359	393	250	361	200
460A-5	R11	404	660	460	315	398	450	315	414	250
503A-5	R11	455	725	503	355	436	492	355	460	315
U_n = 690 V										
142A-7	R11	123	250	142	132	170	135	132	119	110
174A-7	R11	149	274	174	160	208	165	160	142	132
210A-7	R11	186	384	210	200	251	200	200	174	160
271A-7	R11	232	411	271	250	324	257	250	210	200
330A-7	R11	293	480	330	315	394	320	315	271	250
370A-7	R11	330	520	370	355	442	360	355	330	315
430A-7	R11	375	555	430	400	514	420	400	370	355

UL (NEC) RATINGS								
ACS880-34-...	**Frame size**	**Input current[1)]**	**Output ratings**					
			Max. current	**App. power**	**Light-duty use**		**Heavy-duty use**	
		I_1	I_{max}	S_n	I_{Ld}	P_{Ld}	I_{Hd}	P_{Hd}
		A	**A**	**kVA**	**A**	**hp**	**A**	**hp**
U_n = 480 V								
240A-5	R11	169	350	208	240	200	228	150
260A-5	R11	205	418	225	260	200	240	200
302A-5	R11	239	498	262	302	250	260	200
361A-5	R11	257	542	313	361	300	302	250
414A-5	R11	321	614	359	414	350	361	300
460A-5	R11	404	660	398	450	350	414	350
503A-5	R11	455	725	436	492	400	483	400
U_n = 575 V								
142A-7	R11	123	250	170	144	150	125	125
174A-7	R11	149	274	208	168	175	144	150
210A-7	R11	186	384	251	192	200	168	175
271A-7	R11	232	411	324	242	250	192	200
330A-7	R11	293	480	394	289	300	242	250
370A-7	R11	330	520	442	336	350	289	300
430A-7	R11	375	555	514	412	450	336	350

[1)] When the DC voltage is boosted, the drive can draw more input current than shown on the type designation label. This is the case when the motor is running continuously at or near the field weakening area and when the drive is running at or near nominal load. It can be a result of certain combinations of DC voltage boost levels and drive-type-specific derating curves.

The rise in input current can heat the input cable and fuses. To avoid heating, select an input cable and fuses according to the increased input current caused by the DC voltage boost. For more information, refer to ACS880-11, ACS880-31, ACS880-14, ACS880-34, ACS880-17, ACS880-37 drives product note on voltage boost (3AXD50000691838 [English]).

U_n	Nominal voltage of the drive
I_1	Nominal input current (rms) at 40 °C (104 °F)
S_n	Apparent power (no overload)
I_{max}	Maximum output current. Available for 10 seconds at start, otherwise as long as allowed by drive temperature. 140% … 200% of I_{Hd}, depending on power rating.
I_2	Continuous rms output current. No overload capability at 40 °C (104 °F). This is indicated in the type designation label as output current I_2.
P_n	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
P_{Ld}	Typical motor power for light-duty use
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes
P_{Hd}	Typical motor power for heavy-duty use
Note: To achieve the rated motor power given in the table, the rated current of the drive must be higher than or equal to the rated motor current. The power ratings apply to most IEC 34 motors at the nominal voltage of the drive. ABB recommends to select the drive, motor and gear combination for the required motion profile with the DriveSize dimensioning tool available from ABB.	

Deratings

When is derating necessary

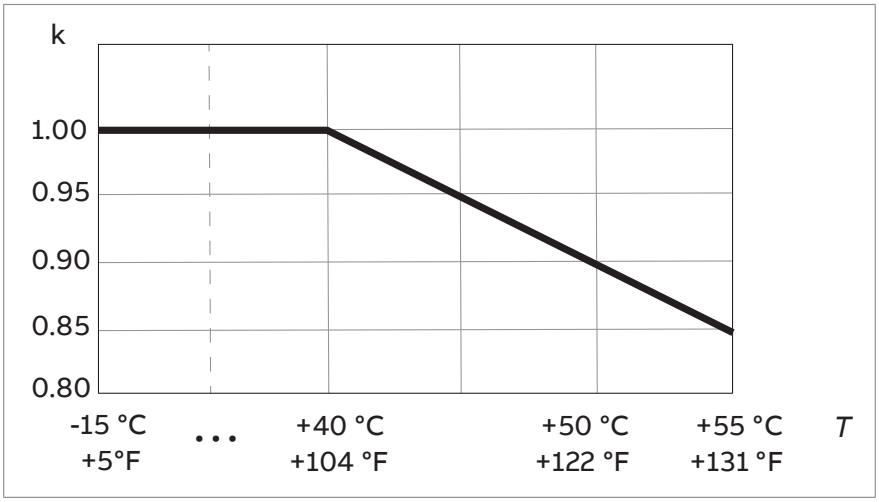
Derate the continuous output current of the drive if

- ambient temperature exceeds +40 °C (+104 °F) or
- drive is installed higher than 1000 m (3280 ft) above sea level
- switching frequency is other than default
- the minimum requirements of motor cable length are not met (see chapter d*u*/d*t* filters and sine filters)
- DC voltage boost feature is used.

Note: The final derating factor is a multiplication of all applicable derating factors.

Surrounding air temperature derating

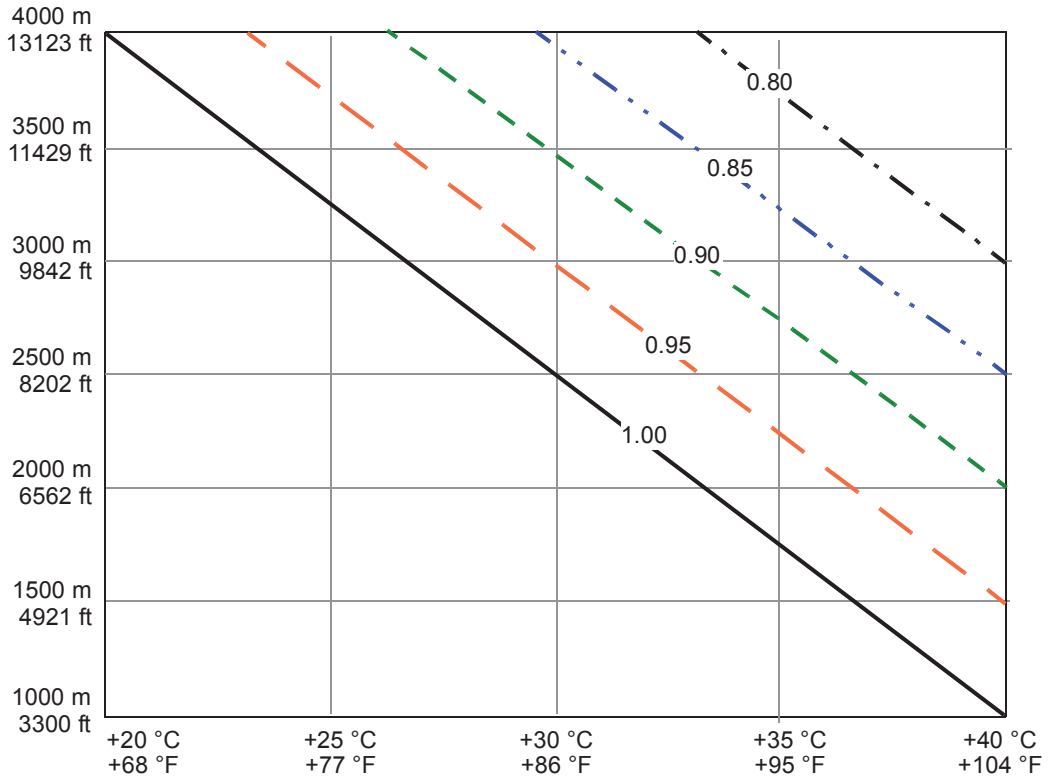
In the temperature range +40…55 °C (+104…131 °F), the rated output current is derated by 1% for every added 1 °C (1.8 °F) as follows. Calculate the output current by multiplying the current given in the rating table by the derating factor.



Altitude derating

At altitudes more than 1000 m (3281 ft) above sea level, the output current derating is 1 percentage point for every added 100 m (328 ft). For example, the derating factor for 1500 m (4921 ft) is 0.95. The maximum permitted installation altitude is given in the technical data.

If the surrounding air temperature is less than +40 °C (104 °F), the derating can be reduced by 1.5 percentage points for every 1 °C (1.8 °F) reduction in temperature. A few altitude derating curves are shown below.



	Derating factor 1.00
	Derating factor 0.95
	Derating factor 0.90
	Derating factor 0.85
	Derating factor 0.80

For a more accurate derating, use the DriveSize PC tool.

Deratings for special settings in the drive control program

Enabling special settings in the drive control program can require output current derating.

Ex motor, sine filter, low noise

Table below gives the deratings for these cases:

- drive is used with an ABB motor for explosive atmospheres (Ex) and **EX motor** in parameter 95.15 Special HW settings is enabled
- sine filter given in the selection table (see section Sine filters) is used and **ABB sine filter** in parameter 95.15 Special HW settings is enabled
- **Low noise optimization** is selected in parameter 97.09 Switching freq mode.

With other than recommended sine filters (see section Sine filters (Page 256)) and non-ABB Ex motors, contact ABB.

ACS880-34-...	Output ratings for special settings											
	Ex motor (ABB Ex motor)				ABB sine filter				Low noise mode			
	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use
	I_n	P_n	I_{Ld}	I_{Hd}	I_n	P_n	I_{Ld}	I_{Hd}	I_n	P_n	I_{Ld}	I_{Hd}
	A	kW	A	A	A	kW	A	A	A	kW	A	A
U_n = 400 V												
246A-3	234	132	222	196	221	132	210	185	217	132	204	180
293A-3	278	160	264	234	264	160	251	221	258	160	243	215
363A-3	345	200	328	278	327	200	310	264	320	200	301	256
442A-3	420	250	399	345	398	250	378	327	390	250	367	317
505A-3	480	315	456	345	455	250	432	327	445	250	419	317
585A-3	556	315	528	420	527	315	500	398	516	315	485	386
650A-3	618	355	587	480	585	355	556	455	573	315	539	441
U_n = 480 V												
302A-5	287	250 hp	287	247	272	250 hp	272	234	266	250 hp	264	277
U_n = 500 V												
240A-5	228	132	217	171	216	132	205	162	212	132	199	157
260A-5	247	160	235	228	234	160	222	216	229	160	216	210
361A-5	343	200	326	247	325	200	309	234	318	200	300	227
414A-5	393	250	373	343	373	250	354	325	365	250	343	315
460A-5	437	315	428	393	414	315	405	373	406	250	393	362
503A-5	478	355	467	437	453	315	443	414	443	315	430	402
U_n = 690 V												
142A-7	125	132	119	105	128	132	122	107	66	75	63	55
174A-7	153	160	145	125	157	160	149	128	81	90	77	66
210A-7	185	200	176	153	189	200	180	157	98	110	93	81
271A-7	238	250	226	185	244	250	231	189	126	132	119	98
330A-7	290	315	282	238	297	315	288	244	154	160	149	126

ACS880-34-...	Output ratings for special settings											
	Ex motor (ABB Ex motor)				ABB sine filter				Low noise mode			
	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use	Nominal use		Light-duty use	Heavy-duty use
	I_n	P_n	I_{Ld}	I_{Hd}	I_n	P_n	I_{Ld}	I_{Hd}	I_n	P_n	I_{Ld}	I_{Hd}
	A	kW	A	A	A	kW	A	A	A	kW	A	A
370A-7	326	355	317	290	333	355	324	297	172	200	167	153
430A-7	378	400	370	326	387	400	378	333	200	200	195	172

U_n	Nominal voltage of the drive
I_n	Continuous rms output current. No overload capability at 40 °C (104 °F)
P_n	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
P_{Ld}	Typical motor power for light-duty use
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes

High speed mode

The selection **High speed mode** of parameter 95.15 Special HW settings improves control performance at high output frequencies. ABB recommends it to be selected with output frequency of 120 Hz and above.

This table gives the drive module ratings for 120 Hz output frequency and the maximum output frequency for the drive ratings when **High speed mode** in parameter 95.15 Special HW settings is enabled: With output frequencies smaller than this recommended maximum output frequency, the current derating is less than the values given in the table. Contact ABB for operation above the recommended maximum output frequency or for the output current derating with output frequencies above 120 Hz and below the maximum output frequency.

ACS880-34-	Deratings with selection High speed mode of parameter 95.15 Special HW settings									
	120 Hz output frequency					Maximum output frequency				
	Frequency	Nominal use		Light-duty use	Heavy-duty use	Maximum frequency	Nominal use		Light-duty use	Heavy-duty use
	f	I_n	P_n	I_{Ld}	I_{Hd}	f_{max}	I_n	P_n	I_{Ld}	I_{Hd}
	Hz	A	kW	A	A	Hz	A	kW	A	A
U_n = 400 V										
246A-3	120	246	132	234	206	500	201	110	193	170
293A-3	120	293	160	278	246	500	240	132	229	203
363A-3	120	363	200	345	293	500	297	200	284	241
442A-3	120	442	250	420	363	500	362	250	346	299
505A-3	120	505	250	480	363	500	413	250	395	299
582A-3	120	585	315	556	442	500	479	315	458	364
650A-3	120	650	355	618	505	500	532	315	509	416
U_n = 480 V										
302A-5	120	302	250 (hp)	302	260	500	247	200 (hp)	249	214

ACS880-34-	Deratings with selection High speed mode of parameter 95.15 Special HW settings									
	120 Hz output frequency					Maximum output frequency				
	Frequency	Nominal use		Light-duty use	Heavy-duty use	Maximum frequency	Nominal use		Light-duty use	Heavy-duty use
	f	I_n	P_n	I_{Ld}	I_{Hd}	f_{max}	I_n	P_n	I_{Ld}	I_{Hd}
	Hz	A	kW	A	A	Hz	A	kW	A	A
U_n = 500 V										
240A-5	120	240	132	228	180	500	196	132	188	148
260A-5	120	260	160	247	240	500	213	160	203	198
361A-5	120	361	200	343	260	500	295	250	283	214
414A-5	120	414	250	393	361	500	339	250	324	297
460A-5	120	460	315	450	414	500	376	315	371	341
503A-5	120	503	355	492	460	500	412	315	405	379
U_n = 690 V										
142A-7	120	142	132	135	119	500	82	75	78	68
174A-7	120	174	160	165	142	500	100	110	95	82
210A-7	120	210	200	200	174	500	121	132	115	100
271A-7	120	271	250	257	210	500	156	160	148	121
330A-7	120	330	315	320	271	500	190	200	184	156
370A-7	120	370	355	360	330	500	213	250	207	190
430A-7	120	430	400	420	370	500	247	250	241	213
										3AXD00000588487

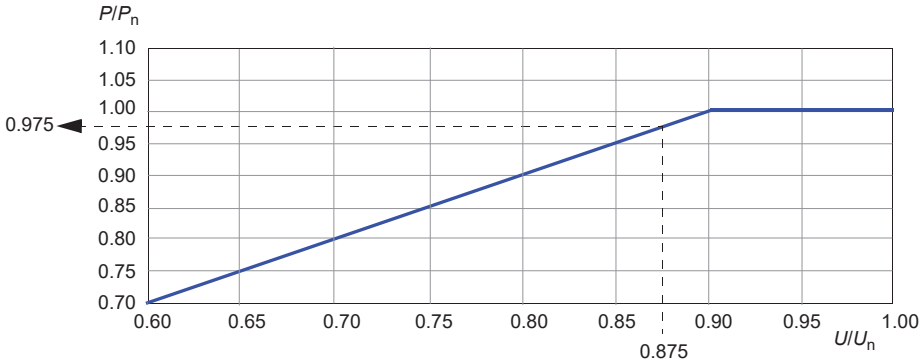
f	Output frequency
f_{max}	Maximum output frequency with High speed mode
U_n	Nominal voltage of the drive
I_n	Continuous rms output current. No overload capability at 40 °C (104 °F).
P_n	Typical motor power in no-overload use
I_{Ld}	Continuous rms output current allowing 10% overload for 1 minute every 5 minutes
I_{Hd}	Continuous rms output current allowing 50% overload for 1 minute every 5 minutes

Derating for output voltage boost

The drive can output a higher motor voltage than the supply voltage. This can require derating of the drive output power depending on the difference between the supply voltage and the output voltage to the motor for continuous operation.

400 V and 500 V drives

This drawing shows the required derating for -3 and -5 (400 V and 500 V) drive types.



Example 1: P_n for ACS880-34-650A-3 is 355 kW. The input voltage (U) is 350 V. -> U/U_n = 350 V / 400 V = 0.875. -> P/P_n = 0.975 -> The derated power P = 0.975 × 355 kW = 346 kW.

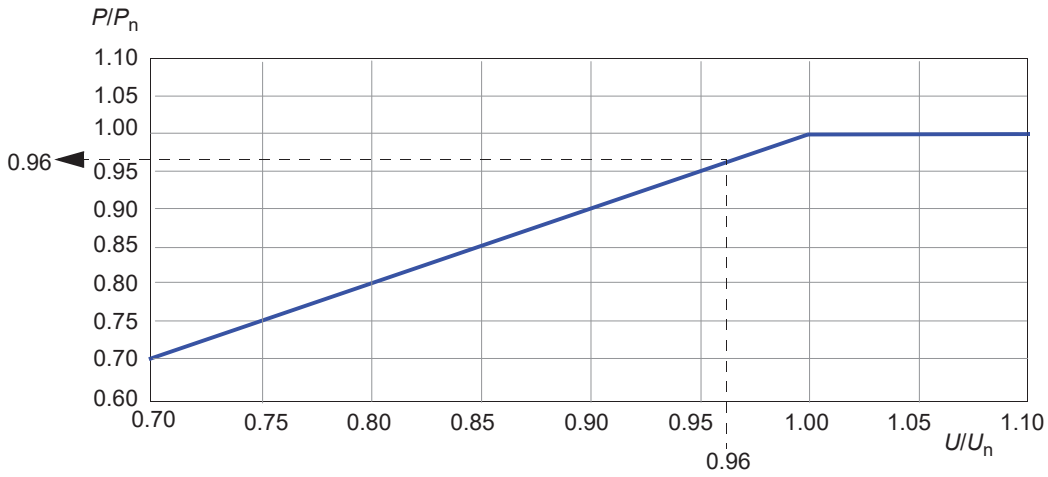
To boost the output voltage to correspond the nominal supply voltage 400 V, increase the DC voltage to 400 V × √2 = 567 V.

Example 2: P_n for ACS880-34-503A-5 is 355 kW. The input voltage (U) is 450 V. -> U/U_n = 450 V / 500 V = 0.9. -> P/P_n = 1.00 -> The derated power P = 1.00 × 355 kW = 355 kW.

To boost the output voltage to correspond the nominal supply voltage 500 V, increase the DC voltage to 500 V × √2 = 707 V.

575 V and 690 V drives

This drawing shows the required derating for -7 (575 V and 690 V) drive types.



Example 1: P_n for ACS880-34-430A-7 is 400 kW. The input voltage (U) is 660 V. -> U/U_n = 660 V / 690 V = 0.96 -> P/P_n = 0.96 -> The derated power P = 0.96 × 400 kW = 384 kW.

To boost the output voltage to correspond the nominal supply voltage 690 V, increase the DC voltage to 690 V × √2 = 977 V.

U	Input voltage of the drive
U_n	Nominal supply voltage of the drive. For -3 types U_n = 400 V, for -5 types U_n = 500 V. For -7 types U_n = 690 V but 575 V when P_n refers to nominal power ratings in the UL (NEC) 575 V rating table.
P	Derated output power of the drive
P_n	Nominal power rating of the drive

For more information, see ACS880-11, ACS880-31, ACS880-14, ACS880-34, ACS880-17, ACS880-37 drives product note on voltage boost (3AXD50000691838 [English]).

Fuses (IEC)

aR fuses by Cooper Bussmann for protection against short-circuit in the input power cable of drive are listed below.

Ultrarapid (aR) fuses per drive module							
ACS880-34-...	Min. short-circuit current [1]	Input current	Fuse				
	A	A	A	A^2s	V	Bussmann type	Size
U_n = 400 V							
246A-3	1500	212	400	74000	690	170M5408	2
293A-3	2200	257	500	145000	690	170M5410	2
363A-3	2600	321	630	210000	690	170M6410	3
442A-3	3100	401	700	300000	690	170M6411	3
505A-3	4000	401	800	465000	690	170M6412	3
585A-3	5400	505	1000	945000	690	170M6414	3
650A-3	5400	569	1000	945000	690	170M6414	3
U_n = 500 V							
240A-5	1100	169	315	42000	690	170M4410	1
260A-5	1500	205	400	74000	690	170M5408	2
361A-5	2600	257	630	210000	690	170M6410	3
414A-5	3100	321	700	300000	690	170M6411	3
460A-5	3100	404	700	300000	690	170M6411	3
503A-5	4000	455	800	465000	690	170M6412	3
U_n = 690 V							
142A-7	900	123	250	21000	690	170M4409	1
174A-7	1100	149	315	42000	690	170M4410	1
210A-7	1500	186	400	74000	690	170M5408	2
271A-7	2200	232	500	145000	690	170M5410	2
330A-7	2600	293	630	210000	690	170M6410	3
370A-7	3100	330	700	300000	690	170M6411	3
430A-7	3100	375	700	300000	690	170M6411	3

[1] Minimum short-circuit current of the electrical power system

Note:

- See also sections:
  - Protecting the drive and the input power cable in short-circuits (Page 85)
  - Protecting the drive against thermal overload (Page 85)
  - Protecting the input power cable against thermal overload (Page 86).
- In multicable installations, install only one fuse per phase (not one fuse per conductor).
- Fuses with higher current rating than the recommended ones must not be used. Fuses with lower current rating can be used.

- Fuses from other manufacturers can be used if they agree with the ratings and the melting curve of the fuse does not exceed the melting curve of the fuse mentioned in the table.

Calculating the short-circuit current of the installation

Make sure that the short-circuit current of the installation is at least the value given in the fuse table.

The short-circuit current of the installation can be calculated as follows:

$$I_{k2\text{-}ph} = \frac{U}{2 \cdot \sqrt{R_c^2 + (Z_k + X_c)^2}}$$

where

$I_{k2\text{-}ph}$	Short-circuit current in symmetrical two-phase short-circuit
U	Network line-to-line voltage (V)
R_c	Cable resistance (ohm)
Z_k	$Z_k = z_k \cdot U_n^2 / S_n$ = transformer impedance (ohm)
z_k	Transformer impedance (%)
U_n	Transformer rated voltage (V)
S_n	Nominal apparent power of the transformer (kVA)
X_c	Cable reactance (ohm)

Fuses (UL)

UL fuses by Cooper Bussman for branch circuit protection per NEC per drive module are listed below. Obey local regulations.

ACS880-34-...	Input current (A)	Fuse					
		A	V	Type DIN 43653	Type US Style	Type French Style	Size
U_n = 500 V							
240A-5	169	315	690	170M4010	170M4610	170M4310	1
260A-5	205	400	690	170M5008	170M5608	170M5308	2
302A-5	249	500	690	170M5010	170M5610	170M5310	2
361A-5	257	630	690	170M6010	170M6610	170M6310	3
414A-5	321	700	690	170M6011	170M6611	170M6311	3
460A-5	404	700	690	170M6011	170M6611	170M6311	3
503A-5	455	800	690	170M6012	170M6612	170M6212	3
U_n = 690 V							
142A-7	123	250	690	170M4009	170M4609	170M4309	1
174A-7	149	315	690	170M4010	170M4610	170M4310	1
210A-7	186	400	690	170M5008	170M5608	170M5308	2
271A-7	232	500	690	170M5010	170M5610	170M5310	2
330A-7	293	630	690	170M6010	170M6610	170M6310	3
370A-7	330	700	690	170M6011	170M6611	170M6311	3
430A-7	375	700	690	170M6011	170M6611	170M6311	3

Note:

- See also sections:
  - Protecting the drive and the input power cable in short-circuits (Page 85)
  - Protecting the drive against thermal overload (Page 85)
  - Protecting the input power cable against thermal overload (Page 86).
- In multicable installations, install only one fuse per phase (not one fuse per conductor).
- Fuses with higher current rating than the recommended ones must not be used. Fuses with lower current rating can be used.
- Alternative fuses can be used if they meet certain characteristics. For acceptable fuses, see the manual supplement (3AXD50000645015).

Circuit breakers (UL)

Note: Circuit breakers must not be used without fuses. For suitable circuit breakers, contact your local ABB representative.

Dimensions, weights and free space requirements

Standard drive module configuration (drive module + LCL filter module) – IP20 (UL Open Type)								
Frame size	Height		Width		Depth		Weight	
	mm	in	mm	in	mm	in	kg	lb
R11	1741	68.54	713	28.07	512	20.16	373	822

Optional selection +0B051+0H371 (without shrouds and full-size output power cable connection terminals) with LCL filter module – IP00 (UL Open Type)								
Frame size	Height		Width		Depth		Weight	
	mm	in	mm	in	mm	in	kg	lb
R11	1726	67.93	642	25.27	508	20.00	365	804

Drive module								
Frame size	Height		Width		Depth		Weight	
	mm	in	mm	in	mm	in	kg	lb
R11	1726	67.93	404	15.92	508	20.00	185	408

LCL filter module								
Frame size	Height		Width		Depth		Weight	
	mm	in	mm	in	mm	in	kg	lb
R11	1722	67.80	239	9.40	505	19.86	180	396

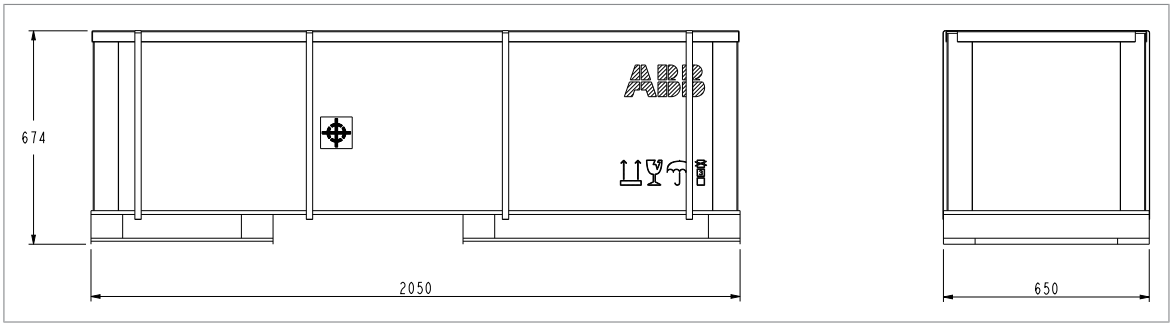
Optional selection +H381 (full power cabling panels) with LCL filter module								
Frame size	Height		Width		Depth		Weight	
	mm	in	mm	in	mm	in	kg	lb
R11	1780	70.08	709	27.91	517	20.35	401	884

Weight of optional selections								
Frame size	+E208		+0H371		+H370		+0B051	
	kg	lb	kg	lb	kg	lb	kg	lb
R11	3	7	-2.9	-6	2.9	6	-1.5	-3

For requirements of free space around the drive module, see Required free space (Page 54).

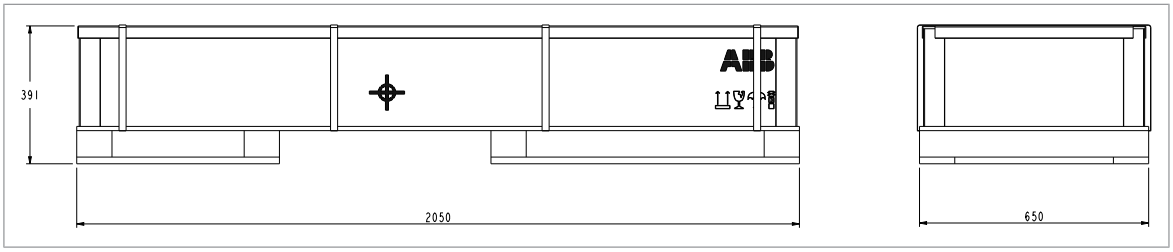
Package

Drive package



Weight: 36 kg (79 lb).

LCL filter module package



Weight: 32 kg (71 lb).

Losses, cooling data and noise

The air flow direction is from bottom to top.

This table shows typical heat loss values, required air flow and noise at the nominal ratings of the drive. The heat loss values can vary depending on voltage, cable conditions, motor efficiency and power factor. To obtain more accurate values for given conditions, use ABB DriveSize tool (http://new.abb.com/drives/software-tools/drivesize).

ACS880-34-	Frame size	Air flow		Heat dissipation	Noise
		m³/h	ft³/min	W	dB(A)
U_n = 400 V					
246A-3	R11	2100	1236	5280	75
293A-3	R11	2100	1236	6400	75
363A-3	R11	2100	1236	8000	75
442A-3	R11	2100	1236	10000	75
505A-3	R11	2100	1236	10000	75
585A-3	R11	2100	1236	12600	75
650A-3	R11	2100	1236	14200	75
U_n = 500 V					
240A-5	R11	2100	1236	5280	75
260A-5	R11	2100	1236	6400	75
302A-5	R11	2100	1236	8000	75
361A-5	R11	2100	1236	8000	75
414A-5	R11	2100	1236	10000	75
460A-5	R11	2100	1236	12600	75
503A-5	R11	2100	1236	14200	75
U_n = 690 V					
142A-7	R11	2100	1236	5280	75
174A-7	R11	2100	1236	6400	75
210A-7	R11	2100	1236	8000	75
271A-7	R11	2100	1236	10000	75
330A-7	R11	2100	1236	12600	75
370A-7	R11	2100	1236	14200	75
430A-7	R11	2100	1236	16000	75

These losses are not calculated according to IEC 61800-9-2.

The cooling air temperature rises 30 °C when it goes through the drive module if the temperature of the input cooling air is 40 °C.